

Practice Quiz: Cognition (Mathematical Frameworks)

Name: __ Date: __

Part A: Multiple Choice

1. A POMDP adds what to an HMM? A) More observations B) Agency — the ability to choose actions that change state transitions C) Faster computation D) Fewer states
2. The A matrix in Active Inference represents: A) $P(s_t | s_{t-1})$ — how states evolve B) $P(o | s)$ — how hidden states generate observations C) The agent's preferences D) The agent's initial beliefs
3. The C vector encodes: A) Transition dynamics B) Which observations the agent prefers or avoids C) The number of hidden states D) The probability of each action
4. In the belief update equation $q(s_t) \propto A(o_t, :) \times B \times q(s_{t-1})$, what does $A(o_t, :)$ do? A) Selects the row of A corresponding to the current observation, providing the likelihood B) Updates the transition model C) Changes the agent's preferences D) Resets the agent's beliefs
5. A policy π in this framework is: A) A single action B) A sequence of actions to be evaluated for their expected outcomes C) A brain region D) A probability distribution over states
6. "Partially observable" means: A) The agent can see everything perfectly B) The agent cannot directly observe hidden states — observations are noisy/indirect C) Half the states don't exist D) The model is incomplete
7. The D vector represents: A) The agent's beliefs about the initial state before any observations B) The discount factor C) The dimensionality of the model D) The agent's actions

Part B: Short Answer

1. Construct an A matrix for a 2-state, 2-observation system where observation is very reliable (90% correct). Write it out and verify columns sum to 1.

2. Explain why the C vector replaces the reward function in Active Inference. What's the philosophical difference between "seeking preferred observations" and "maximizing reward"?
3. Trace one full step of the perception-action loop: Given $D = [0.6, 0.4]$, $A = [[0.9, 0.2], [0.1, 0.8]]$, and observation $o = \text{"obs_1"}$, compute the posterior $q(s)$ (show your work). Which state is most likely?